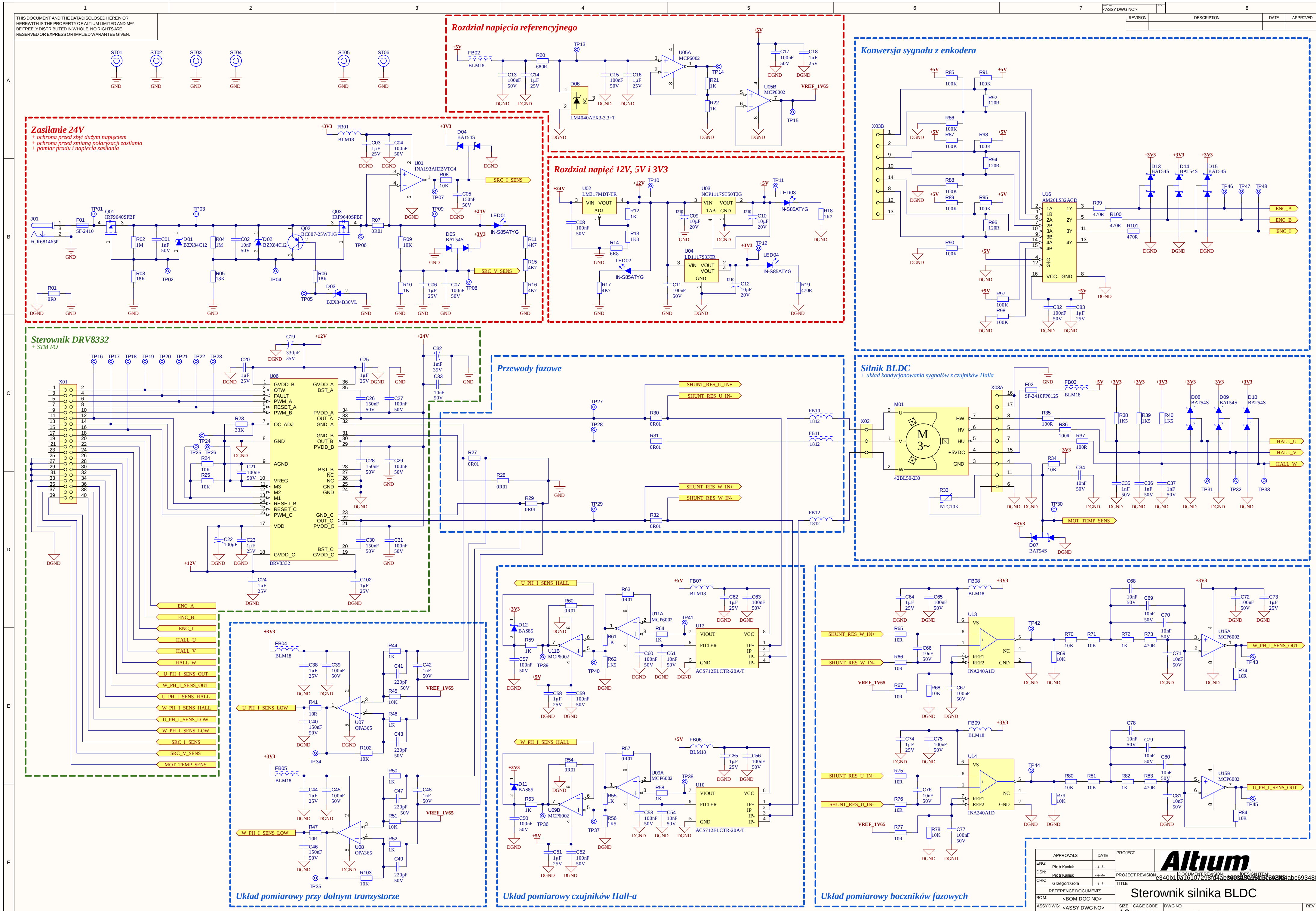
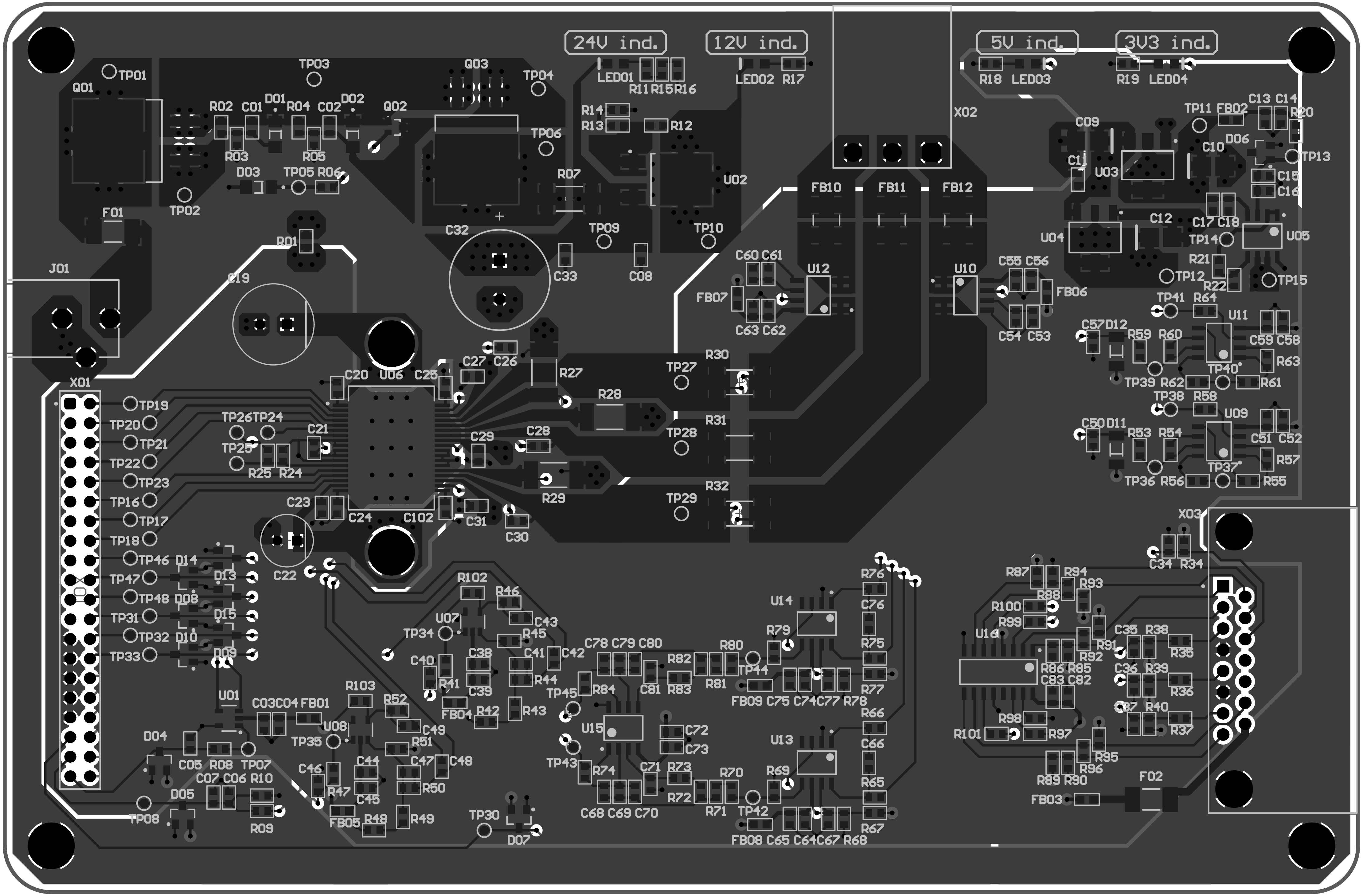
THIS DOCUMENT AND THE DATA DISCLOSED HEREIN OR HEREWITH IS THE PROPERTY OF ALTIUM LIMITED AND MAY BE FREELY DISTRIBUTED IN WHOLE. NO RIGHTS ARE RESERVED OR EXPRESS OR IMPLIED WARANTEE GIVEN.

REVISION	DESCRIPTION	DATE	APPROVED


ST01 ST02 ST03 ST04 ST05 ST06 GND
Rozdział napięcia referencyjnego
FB02 BLM18, R20 680R, C13 100nF 50V, C14 1µF 25V, D06 LM4040AEX3-3.3+T, TP13, C15 100nF 50V, C16 1µF 25V, U05A MCP6002, TP14, R21 1K, R22 1K, U05B MCP6002, C17 100nF 50V, C18 1µF 25V, VREF_1V65, TP15
Zasilanie 24V
+ ochrona przed zbyt dużym napięciem
+ ochrona przed zmianą polaryzacji zasilania
+ pomiar prądu i napięcia zasilania
J01 FCR681465P, F01 SF-2410, TP01, Q01 IRF9640SPBF, R02 1M, C01 1nF 50V, D01 BZX84C12, TP02, R03 18K, R01 0R0, TP03, R04 1M, C02 10nF 50V, D02 BZX84C12, R05 18K, TP04, Q02 BC807-25WT1G, R06 18K, D03 BZX84B30VL, TP05, Q03 IRF9640SPBF, TP06, R07 0R01
FB01 BLM18, C03 1µF 25V, C04 100nF 50V, U01 INA193AIDBVTG4, R08 10K, TP07, C05 150nF 50V, D04 BAT54S, SRC_I_SENS, TP09, LED01 IN-S85ATYG, D05 BAT54S, R09 10K, R10 1K, C06 1µF 25V, C07 100nF 50V, TP08, SRC_V_SENS, R11 4K7, R15 4K7, R16 4K7
Rozdział napięć 12V, 5V i 3V3
U02 LM317MDT-TR, C08 100nF 50V, R12 1K, R13 1K8, R14 6K8, LED02 IN-S85ATYG, R17 4K7, TP10, U03 NCP1117ST50T3G, C09 10µF 20V, C10 10µF 20V, TP11, LED03 IN-S85ATYG, R18 1K2, U04 LD1117S33TR, C11 100nF 50V, C12 10µF 20V, TP12, LED04 IN-S85ATYG, R19 470R
Konwersja sygnału z enkodera
X03B, R85 100K, R86 100K, R87 100K, R88 100K, R89 100K, R90 100K, R91 100K, R92 120R, R93 100K, R94 120R, R95 100K, R96 120R, R97 100K, R98 100K, U16 AM26LS32ACD, C82 100nF 50V, C83 1µF 25V, R99 470R, R100 470R, R101 470R, D13 BAT54S, D14 BAT54S, D15 BAT54S, TP46, TP47, TP48, ENC_A, ENC_B, ENC_I
Sterownik DRV8332
+ STM I/O
X01, TP16 TP17 TP18 TP19 TP20 TP21 TP22 TP23, C19 330µF 35V, C20 1µF 25V, C25 1µF 25V, U06 DRV8332, R23 33K, C26 150nF 50V, C27 100nF 50V, TP24, TP25, TP26, R24 10K, R25 10K, C21 100nF 50V, C28 150nF 50V, C29 100nF 50V, C22 100µF, C23 1µF 25V, C30 150nF 50V, C31 100nF 50V, C24 1µF 25V, C102 1µF 25V, C32 1mF 35V, C33 10nF 50V
ENC_A, ENC_B, ENC_I, HALL_U, HALL_V, HALL_W, U_PH_I_SENS_OUT, W_PH_I_SENS_OUT, U_PH_I_SENS_HALL, W_PH_I_SENS_HALL, U_PH_I_SENS_LOW, W_PH_I_SENS_LOW, SRC_I_SENS, SRC_V_SENS, MOT_TEMP_SENS
Przewody fazowe
SHUNT_RES_U_IN+, SHUNT_RES_U_IN-, SHUNT_RES_W_IN+, SHUNT_RES_W_IN-, TP27, TP28, TP29, R27 0R01, R28 0R01, R29 0R01, R30 0R01, R31 0R01, R32 0R01, FB10 1812, FB11 1812, FB12 1812
Silnik BLDC
+ układ kondycjonowania sygnałów z czujników Halla
X02, M01 42BL50-230, X03A, F02 SF-2410FP0125, FB03 BLM18, R35 100R, R36 100R, R37 100R, R34 10K, C34 10nF 50V, R33 NTC10K, TP30, D07 BAT54S, R38 1K5, R39 1K5, R40 1K5, D08 BAT54S, D09 BAT54S, D10 BAT54S, C35 1nF 50V, C36 1nF 50V, C37 1nF 50V, TP31, TP32, TP33
Układ pomiarowy przy dolnym tranzystorze
FB04 BLM18, C38 1µF 25V, C39 100nF 50V, R44 1K, C41 220pF 50V, C42 1nF 50V, R45 10K, R41 10R, C40 150nF 50V, U07 OPA365, R46 1K, C43 220pF 50V, R102 10K, TP34, FB05 BLM18, C44 1µF 25V, C45 100nF 50V, R50 1K, C47 220pF 50V, C48 1nF 50V, R51 10K, R47 10R, C46 150nF 50V, U08 OPA365, R52 1K, C49 220pF 50V, R103 10K, TP35
Układ pomiarowy czujników Hall-a
D12 BAS85, R60 0R01, R59 1K, U11B MCP6002, C57 100nF 50V, TP39, C58 1µF 25V, C59 100nF 50V, TP40, R63 0R01, U11A MCP6002, R61 1K, R62 1K5, R64 1K, TP41, FB07 BLM18, C62 1µF 25V, C63 100nF 50V, C60 100nF 50V, C61 10nF 50V, U12 ACS712ELCTR-20A-T
D11 BAS85, R54 0R01, R53 1K, U09B MCP6002, C50 100nF 50V, TP36, C51 1µF 25V, C52 100nF 50V, TP37, R57 0R01, U09A MCP6002, R55 1K, R56 1K5, R58 1K, TP38, FB06 BLM18, C55 1µF 25V, C56 100nF 50V, C53 100nF 50V, C54 10nF 50V, U10 ACS712ELCTR-20A-T
Układ pomiarowy boczników fazowych
FB08 BLM18, C64 1µF 25V, C65 100nF 50V, U13 INA240A1D, R65 10R, R66 10R, C66 10nF 50V, R67 10R, R68 10K, C67 100nF 50V, TP42, R69 10K, R70 10K, R71 10K, R72 1K, R73 470R, C68 10nF 50V, C69 10nF 50V, C70 10nF 50V, C71 10nF 50V, U15A MCP6002, C72 100nF 50V, C73 1µF 25V, R74 10R, TP43
FB09 BLM18, C74 1µF 25V, C75 100nF 50V, U14 INA240A1D, R75 10R, R76 10R, C76 10nF 50V, R77 10R, R78 10K, C77 100nF 50V, TP44, R79 10K, R80 10K, R81 10K, R82 1K, R83 470R, C78 10nF 50V, C79 10nF 50V, C80 10nF 50V, C81 10nF 50V, U15B MCP6002, R84 10R, TP45
+24V, +12V, +5V, +3V3, DGND, GND

APPROVALS	DATE	PROJECT	
ENG: Piotr Kaniuk	--/--/--		
DSN: Piotr Kaniuk	--/--/--	PROJECT REVISION: e340b19a16107298fd4abc69348095fd6e542	DOCUMENT REVISION: DESIGN ITEM:
CHK: Grzegorz Góra	--/--/--	TITLE	
REFERENCE DOCUMENTS		Sterownik silnika BLDC	
BOM: <BOM DOC NO>			
ASSY DWG: <ASSY DWG NO>		SIZE A2	CAGE CODE ?????
FAB DWG: <FAB DWG NO>		DWG NO. <SCH DWG NO>	REV
PCB DWG: <PCB DWG NO>		SCALE:	FILE NAME Sheet1.SchDoc

SHEET 1 OF


24V ind.
12V ind.
5V ind.
3V3 ind.
LED01
R11 R15 R16
LED02
R17
R18
LED03
R19
LED04
TP01
Q01
TP03
Q03
TP04
R02
C01
D01
R04
C02
D02
Q02
R03
R05
TP06
D03
TP05
R06
TP02
F01
R14
R13
R12
R07
U02
X02
TP11 FB02
C13 C14
R20
C09
C10
D06
TP13
C11
U03
C15
C16
FB10
FB11
FB12
TP09
TP10
C32
C33
C08
R01
C12
C17 C18
U04
TP14
U05
C60 C61
U12
U10
C55 C56
R21
TP12
TP15
R22
J01
C19
FB07
FB06
TP41
R64
C63 C62
C54 C53
U11
C57 D12
R59
R60
C59 C58
R63
C27
C26
R27
R30
TP27
TP39
R62
TP40
R61
X01
C20
U06
C25
R28
TP38
R58
U09
TP19
TP20
TP21
TP22
TP23
TP16
TP17
TP18
TP26 TP24
TP25
R25 R24
C21
C29
C28
R31
TP28
C50 D11
R53
R54
C51 C52
R57
TP37
TP36
R56
R55
R29
R32
TP29
C23
C24
C102
C31
C30
X03
C34
R34
TP46
D14
TP47
D13
TP48
D08
TP31
D15
TP32
D10
TP33
D09
C22
R102
R46
C43
R45
U07
TP34
C38
C41
C42
C40
R41
C39
R44
R76
C76
U14
R87
R94
R93
R88
R100
R99
U16
C35
R38
R91
R92
R35
R86 R85
C83 C82
C36
R39
R36
C78 C79 C80
R80
R82
R81
TP44
R79
R75
R77
C81
R83
FB09
C75 C74 C77 R78
TP45
R84
C87
R40
R37
U01
C03 C04 FB01
R103
R52
U08
C49
FB04
R42
R43
R98
R101
R97
U15
C72
C73
R66
C66
U13
D04
C05
R08
TP07
TP35
R51
C46
C44
C47
C48
R47
C45
R50
TP43
R74
C71
R73
R70
R69
R65
R95
R96
R89 R90
F02
FB03
C07 C06 R10
D05
TP08
R09
FB05
R48
R49
TP30
D07
C68 C69 C70
R72
R71
TP42
FB08 C65 C64 C67 R68
R67

Board Stack Report